

PRACTICAL 4

```
mysql> USE sec_13;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| sec_13      |
+-----+
1 row in set (0.00 sec)

mysql> CREATE TABLE class (
    ->     id INT,
    ->     name VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE class_info (
    ->     id INT,
    ->     address VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_sec_13 |
+-----+
| class             |
| class_info        |
| student           |
| student_marks1    |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO class_info VALUES
-> (1,'DELHI'),
-> (2,'MUMBAI'),
-> (3,'CHENNAI');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT *
-> FROM class
-> CROSS JOIN class_info;
+-----+-----+-----+-----+
| id  | name | id  | address |
+-----+-----+-----+-----+
| 4   | alex | 1   | DELHI   |
| 2   | adam | 1   | DELHI   |
| 1   | abhi | 1   | DELHI   |
| 4   | alex | 2   | MUMBAI  |
| 2   | adam | 2   | MUMBAI  |
| 1   | abhi | 2   | MUMBAI  |
| 4   | alex | 3   | CHENNAI |
| 2   | adam | 3   | CHENNAI |
| 1   | abhi | 3   | CHENNAI |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

```
mysql> INSERT INTO class VALUES
-> (1,'abhi'),
-> (2,'adam'),
-> (3,'alex'),
-> (4,'anu');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> INSERT INTO class_info VALUES
-> (1,'DELHI'),
-> (2,'MUMBAI'),
-> (3,'CHENNAI');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT *
-> FROM class
-> INNER JOIN class_info
-> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
3	alex	3	CHENNAI

10 rows in set (0.00 sec)

```
mysql> SELECT class.name,
-> class_info.address
-> FROM class
-> INNER JOIN class_info
-> ON class.id=class_info.id;
```

name	address
abhi	DELHI
abhi	DELHI
adam	MUMBAI
adam	MUMBAI
abhi	DELHI
abhi	DELHI
adam	MUMBAI
adam	MUMBAI
alex	CHENNAI
alex	CHENNAI

10 rows in set (0.00 sec)

```
mysql> SELECT *  
      -> FROM class  
      -> NATURAL JOIN class_info;
```

id	name	address
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
3	alex	CHENNAI
3	alex	CHENNAI

```
10 rows in set (0.00 sec)
```

```
mysql> INSERT INTO class VALUES  
      -> (5,'ashish');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql>  
mysql> INSERT INTO class_info VALUES  
      -> (7,'NOIDA'),  
      -> (8,'PANIPAT');  
Query OK, 2 rows affected (0.01 sec)  
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT *
-> FROM class
-> LEFT OUTER JOIN class_info
-> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
4	alex	NULL	NULL
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL

13 rows in set (0.00 sec)

```
mysql>
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id=class_info.id
-> WHERE class_info.id IS NULL;
```

id	name	id	address
4	alex	NULL	NULL
4	anu	NULL	NULL
5	ashish	NULL	NULL

3 rows in set (0.00 sec)

```
mysql> SELECT *
      -> FROM class
      -> RIGHT OUTER JOIN class_info
      -> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

12 rows in set (0.00 sec)

```
mysql>
mysql> SELECT *
      -> FROM class
      -> RIGHT JOIN class_info
      -> ON class.id=class_info.id
      -> WHERE class.id IS NULL;
```

id	name	id	address
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

2 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
->
-> UNION
->
-> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
4	alex	NULL	NULL
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

8 rows in set (0.01 sec)

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class_info.id IS NULL
->
-> UNION
->
-> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class.id IS NULL;
```

id	name	id	address
4	alex	NULL	NULL
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

5 rows in set (0.00 sec)


```
mysql> CREATE TABLE first_table(  
-> id INT,  
-> name VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql>  
mysql> CREATE TABLE second_table(  
-> id INT,  
-> name VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql>  
mysql> INSERT INTO first_table VALUES  
-> (1, 'abhi'),  
-> (2, 'adam');  
Query OK, 2 rows affected (0.01 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql>  
mysql> INSERT INTO second_table VALUES  
-> (2, 'adam'),  
-> (3, 'chester');  
Query OK, 2 rows affected (0.01 sec)  
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM first_table
-> UNION
-> SELECT * FROM second_table;
```

id	name
1	abhi
2	adam
3	chester

3 rows in set (0.00 sec)

```
mysql>
mysql> SELECT name FROM first_table
-> UNION
-> SELECT name FROM second_table;
```

name
abhi
adam
chester

3 rows in set (0.00 sec)

```
mysql> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table;
```

id	name
1	abhi
2	adam
2	adam
3	chester

4 rows in set (0.00 sec)

```
mysql> SELECT COUNT(*)
-> FROM
-> (
-> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table
-> ) A;
```

COUNT(*)
4

1 row in set (0.01 sec)

```
mysql> SELECT * FROM first_table
-> INTERSECT
-> SELECT * FROM second_table;
```

id	name
2	adam

1 row in set (0.00 sec)

```
mysql> SELECT name FROM first_table
-> INTERSECT
-> SELECT name FROM second_table;
```

name
adam

1 row in set (0.00 sec)

```
mysql> SELECT f.*
-> FROM first_table AS f
-> LEFT JOIN second_table AS s
-> ON f.id = s.id
-> WHERE s.id IS NULL;
```

```
+-----+-----+
```

```
| id    | name |
```

```
+-----+-----+
```

```
|      1 | abhi |
```

```
+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT name
-> FROM first_table
-> WHERE name NOT IN (
->     SELECT name
->     FROM second_table
-> );
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| abhi |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT c.id,c.name,ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id=ci.id;
```

id	name	address
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
3	alex	CHENNAI
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
3	alex	CHENNAI

10 rows in set (0.00 sec)

```
mysql>
mysql> SELECT c.id,
-> c.name,
-> CASE
-> WHEN ci.address IS NULL
-> THEN 'Address Missing'
-> ELSE 'Address Available'
-> END AS Status
-> FROM class c
-> LEFT JOIN class_info ci
-> ON c.id=ci.id;
```

id	name	Status
1	abhi	Address Available
1	abhi	Address Available
2	adam	Address Available
2	adam	Address Available
4	alex	Address Missing
1	abhi	Address Available
1	abhi	Address Available
2	adam	Address Available
2	adam	Address Available
3	alex	Address Available
3	alex	Address Available
4	anu	Address Missing
5	ashish	Address Missing

13 rows in set (0.00 sec)